

COMPREHENSIVE QUANTUM COMPUTING ULTIMATE EXAM-READY STUDY GUIDE

Complete Coverage of All Curriculum Units with Derivations,
Proofs & Solutions

Based on: Nielsen & Chuang, Rieffel & Polak, Susskind & Friedman
and Your Laboratory Manual

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Contents

1	Introduction to Quantum Computing	5
1.1	History and Motivation	5
1.1.1	Fundamental Concepts	5
1.1.2	Historical Timeline	5
1.1.3	Motivations for Quantum Computing	6
1.2	Differences Between Classical and Quantum Computing	6
1.2.1	Computational Basis and State Notation	7
1.3	Applications of Quantum Computing	7
1.3.1	1. Cryptography and Security	7
1.3.2	2. Drug Discovery and Molecular Simulation	8
1.3.3	3. Optimization Problems	8
1.3.4	4. Machine Learning	9
1.3.5	5. Financial Modeling	9
2	Mathematical Foundations	10
2.1	Complex Numbers	10
2.1.1	Definition and Representation	10
2.1.2	Complex Conjugate	10
2.1.3	Euler's Formula (CRUCIAL)	10
2.1.4	Applications in Quantum Computing	11
2.2	Linear Algebra	11
2.2.1	Vectors in Quantum Computing	11
2.2.2	Inner Product and Orthonormality	12
2.2.3	Matrices and Quantum Gates	12
2.2.4	Hermitian Matrices	12
2.2.5	Unitary Matrices	12
2.2.6	Eigenvalues and Eigenvectors	13
2.2.7	Tensor Product (Kronecker Product)	13
2.3	Dirac Notation (Bra-Ket Notation)	14
2.3.1	Definitions	14
2.3.2	Useful Identities	14
2.3.3	Common Bases	14

3	Quantum Mechanics Basics	15
3.1	The Five Postulates of Quantum Mechanics	15
3.1.1	Postulate 1: State Space	15
3.1.2	Postulate 2: Observables and Measurement Operators . .	16
3.1.3	Postulate 3: Measurement and Collapse	16
3.1.4	Postulate 4: Time Evolution	17
3.1.5	Postulate 5: Composite Systems	17
3.2	Quantum States and Qubits	18
3.2.1	The Bloch Sphere	18
3.3	Superposition	18
3.3.1	Definition	18
3.3.2	Creating Superposition - Hadamard Gate	19
3.3.3	Exponential Scaling	19
3.4	Entanglement	20
3.4.1	Definition	20
3.4.2	Separable vs Entangled States	20
3.4.3	Bell States (Four Maximally Entangled States)	21
3.4.4	Creating Bell States	21
3.4.5	Non-Local Correlations	22
4	Quantum Gates and Circuits	23
4.1	Single-Qubit Gates	23
4.1.1	The Pauli Matrices	23
4.1.2	The Hadamard Gate	25
4.1.3	Phase Gates	26
4.1.4	Rotation Gates	28
4.1.5	Important Gate Relationships and Identities	29
5	Quantum Algorithms (Introduction)	31
5.1	Deutsch's Algorithm	31
5.1.1	Problem Definition	31
5.1.2	Classical Solution	31
5.1.3	Deutsch's Quantum Solution	32
5.1.4	Complete Circuit	34
5.1.5	Quantum Advantage	34
5.2	Quantum Parallelism	34
5.2.1	Concept	34
5.2.2	Mathematical Formulation	34
5.2.3	Example: 3-Qubit Register	34
5.2.4	Classical vs Quantum Computational Cost	35
5.2.5	The Measurement Problem	35
5.3	Exam Practice Questions with Solutions	36
5.3.1	Question 1: Gate Properties (5 marks)	36

5.3.2	Question 2: Hadamard Properties (6 marks)	37
5.3.3	Question 3: Superposition and Measurement (5 marks) . .	38
5.3.4	Question 4: Bell State Entanglement (6 marks)	39
5.3.5	Question 5: Deutsch's Algorithm (6 marks)	40

Chapter 1

Introduction to Quantum Computing

1.1 History and Motivation

1.1.1 Fundamental Concepts

Definition: Quantum computing is a computational paradigm that leverages quantum mechanical phenomena—superposition, entanglement, and interference—to process information fundamentally differently from classical computers.

The exponential advantage of quantum computing arises from the ability to represent and manipulate 2^n states with n qubits, compared to classical systems that represent only one of 2^n states with n bits at any instant.

Classical vs Quantum Information

A classical n -bit register is in exactly one state $x \in \{0, 1\}^n$ at any given time. Computationally:

$$\text{Classical state space} = \{0, 1\}^n \quad (\text{discrete, finite}) \quad (1.1)$$

A quantum n -qubit register exists in a superposition:

$$|\psi\rangle = \sum_{x \in \{0, 1\}^n} \alpha_x |x\rangle, \quad \text{where} \quad \sum_x |\alpha_x|^2 = 1 \quad (1.2)$$

This single state encodes the amplitudes for all 2^n basis states simultaneously.

1.1.2 Historical Timeline

- **1980:** Yuri Manin proposes quantum computing
- **1982:** Richard Feynman suggests quantum computers for simulating quantum systems

- **1985:** David Deutsch formulates the quantum Turing machine
- **1994:** Peter Shor discovers polynomial-time factoring algorithm
- **1996:** Lov Grover develops quantum search algorithm
- **1998-2001:** First experimental implementations (NMR, trapped ions)
- **2019:** Google claims quantum supremacy with 53-qubit processor

1.1.3 Motivations for Quantum Computing

1. Exponential Speedup for Specific Problems

For factoring N -bit numbers:

$$\text{Classical (RSA-2048): } O(e^{1.9(\log N)^{1/3}(\log \log N)^{2/3}}) \approx \text{millions of years} \quad (1.3)$$

$$\text{Quantum (Shor): } O((\log N)^3) \approx \text{hours} \quad (1.4)$$

2. Simulation of Quantum Systems

Classical simulation of quantum chemistry requires exponential resources. Quantum computers can simulate molecular dynamics in polynomial time, enabling:

- Drug discovery and design
- Catalyst optimization
- Material science breakthroughs

3. Optimization and Machine Learning

Quantum algorithms provide quadratic speedup for unstructured search:

$$\text{Classical search: } O(N), \quad \text{Quantum (Grover): } O(\sqrt{N}) \quad (1.5)$$

1.2 Differences Between Classical and Quantum Computing

Detailed Comparison: Superposition

Classical System: Three bits can be in state $|101\rangle$ (decimal 5). To evaluate f on all 8 inputs:

$$\text{Require 8 separate function evaluations} \quad (1.6)$$

Property	Classical	Quantum
Basic Unit	Bit (0 or 1)	Qubit ($ 0\rangle, 1\rangle$, or superposition)
State Representation	One definite value	Superposition (multiple values)
Gates	Irreversible (AND, OR, NOT)	Unitary, reversible
Information Scaling	n bits $\rightarrow$ 1 of 2^n states	n qubits $\rightarrow$ all 2^n states
Measurement	Non-intrusive, deterministic	Intrusive, probabilistic (collapse)
Key Resource	Logic gates	Superposition + Entanglement

Quantum System: Three qubits in superposition:

$$|\psi\rangle = \frac{1}{\sqrt{8}} \sum_{x=0}^7 |x\rangle = \frac{1}{\sqrt{8}} (|000\rangle + |001\rangle + \dots + |111\rangle) \quad (1.7)$$

All 8 values represented simultaneously! This is quantum parallelism.

1.2.1 Computational Basis and State Notation

The standard computational basis for a single qubit:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.8)$$

A general single-qubit state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (1.9)$$

For two qubits:

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (1.10)$$

1.3 Applications of Quantum Computing

1.3.1 1. Cryptography and Security

Breaking Classical Encryption:

- RSA encryption relies on difficulty of factoring $N = pq$ where p, q are large primes
- Shor's algorithm factors in $O((\log N)^3)$ time

- Current threat: “Harvest now, decrypt later” attacks

Quantum Key Distribution:

- BB84 protocol: Theoretically unbreakable encryption
- Any eavesdropping attempt disturbs quantum states (by measurement postulate)
- Legitimate parties detect eavesdropping with certainty

1.3.2 2. Drug Discovery and Molecular Simulation

Quantum chemistry problems intrinsically require quantum computers:

$$\text{Classical simulation: } O(e^N), \quad \text{Quantum simulation: } O(\text{poly}(N)) \quad (1.11)$$

where N is system size.

Applications:

- Protein folding and structure prediction
- Drug-receptor interaction modeling
- Enzyme mechanism elucidation
- Battery development for EVs

1.3.3 3. Optimization Problems

Quantum Approximate Optimization Algorithm (QAOA):

Finds approximate solutions to combinatorial optimization:

$$\max_x f(x), \quad x \in \{0, 1\}^n \quad (1.12)$$

Applications:

- Portfolio optimization (finance)
- Supply chain optimization (logistics)
- Vehicle routing problem (transportation)
- Maximum cut (MaxCut) problem

1.3.4 4. Machine Learning

Quantum machine learning algorithms provide speedup through:

- Quantum feature spaces (exponentially higher dimensional)
- Quantum kernel methods
- Variational quantum algorithms
- Quantum neural networks

Note: Current NISQ (Noisy Intermediate-Scale Quantum) devices are still being tested for practical advantages in ML.

1.3.5 5. Financial Modeling

- Monte Carlo simulations with quadratic speedup
- Risk analysis and portfolio diversification
- Option pricing and derivative valuation
- Time series analysis

Chapter 2

Mathematical Foundations

2.1 Complex Numbers

2.1.1 Definition and Representation

A complex number is expressed as:

$$z = a + bi, \quad \text{where } a, b \in \mathbb{R}, \quad i = \sqrt{-1} \quad (2.1)$$

Polar Form:

$$z = re^{i\theta} = r(\cos \theta + i \sin \theta) \quad (2.2)$$

where $r = |z| = \sqrt{a^2 + b^2}$ (modulus) and $\theta = \arg(z)$ (argument/phase).

2.1.2 Complex Conjugate

For $z = a + bi$:

$$z^* = a - bi \quad (2.3)$$

Key Properties:

$$z \cdot z^* = |z|^2 \quad (\text{always real and non-negative}) \quad (2.4)$$

$$(z_1 + z_2)^* = z_1^* + z_2^* \quad (2.5)$$

$$(z_1 z_2)^* = z_1^* z_2^* \quad (2.6)$$

$$(z^*)^* = z \quad (2.7)$$

2.1.3 Euler's Formula (CRUCIAL)

Theorem: Euler's Formula relates exponential and trigonometric functions:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (2.8)$$

Proof:

Using Taylor series expansion:

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} = \sum_{n=0}^{\infty} \frac{i^n \theta^n}{n!} \quad (2.9)$$

Separating powers of i : $i^0 = 1, i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, \dots$ (period 4)

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right) \quad (2.10)$$

$$= \cos \theta + i \sin \theta \quad (2.11)$$

2.1.4 Applications in Quantum Computing

Phase Shifts

A quantum state with phase θ :

$$|\psi\rangle = e^{i\theta} |0\rangle = (\cos \theta + i \sin \theta) |0\rangle \quad (2.12)$$

Example: T gate applies phase $e^{i\pi/4}$ to basis state $|1\rangle$:

$$T |1\rangle = e^{i\pi/4} |1\rangle \quad (2.13)$$

Superposition with Phases

$$|\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{i}{\sqrt{2}} |1\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/2} |1\rangle) \quad (2.14)$$

The relative phase $e^{i\pi/2}$ between amplitudes affects interference patterns.

2.2 Linear Algebra

2.2.1 Vectors in Quantum Computing

A quantum state is represented as a column vector in a Hilbert space $\mathbb{C}^{2^n}$:

Single Qubit:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (2.15)$$

Two Qubits:

$$|\psi\rangle = \begin{pmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{pmatrix}, \quad \sum_{i,j} |\alpha_{ij}|^2 = 1 \quad (2.16)$$

2.2.2 Inner Product and Orthonormality

For $|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $|\phi\rangle = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$:

Bra-Ket Notation:

$$\langle\psi|\phi\rangle = \alpha^*\gamma + \beta^*\delta \quad (2.17)$$

Properties:

$$\langle\psi|\phi\rangle^* = \langle\phi|\psi\rangle \quad (\text{complex conjugate symmetry}) \quad (2.18)$$

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 \geq 0 \quad (\text{always non-negative real}) \quad (2.19)$$

$$\langle 0|1\rangle = 0 \quad (\text{orthogonality}) \quad (2.20)$$

$$\langle 0|0\rangle = 1, \quad \langle 1|1\rangle = 1 \quad (\text{normalization}) \quad (2.21)$$

2.2.3 Matrices and Quantum Gates

A quantum gate is represented as a matrix acting on state vectors:

$$U|\psi\rangle = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} u_{11}\alpha + u_{12}\beta \\ u_{21}\alpha + u_{22}\beta \end{pmatrix} \quad (2.22)$$

2.2.4 Hermitian Matrices

Definition: A matrix A is Hermitian if $A = A^\dagger$, where $A^\dagger = (A^*)^T$ (conjugate transpose)

Example - Pauli Z:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Z^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^* = Z \quad (2.23)$$

Properties:

- All eigenvalues are real (measurable quantities)
- Eigenvectors form orthonormal basis
- Used to represent observables (measurements)

2.2.5 Unitary Matrices

Definition: A matrix U is unitary if $U^\dagger U = U U^\dagger = I$

All quantum gates must be unitary (reversible, preserve probability).

Example - Pauli X:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X^\dagger = X \quad (2.24)$$

Proof: that X is unitary:

$$X^\dagger X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad \checkmark \quad (2.25)$$

2.2.6 Eigenvalues and Eigenvectors

Definition: For matrix A and non-zero vector $|v\rangle$, if $A|v\rangle = \lambda|v\rangle$, then λ is an eigenvalue with corresponding eigenvector $|v\rangle$

Example - Pauli Z eigenanalysis:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (2.26)$$

Eigenvalues: $\lambda = 1$ or $\lambda = -1$

$$Z|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (+1)|0\rangle \Rightarrow \text{eigenvalue } +1, \text{ eigenvector } |0\rangle \quad (2.27)$$

$$Z|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = (-1)|1\rangle \Rightarrow \text{eigenvalue } -1, \text{ eigenvector } |1\rangle \quad (2.28)$$

Note: In quantum mechanics, measuring an observable represented by Hermitian operator A returns eigenvalue λ with certain probability, and the state collapses to corresponding eigenvector $|v\rangle$.

2.2.7 Tensor Product (Kronecker Product)

For multi-qubit systems, states combine via tensor product:

$$|0\rangle \otimes |0\rangle = |00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |0\rangle \otimes |1\rangle = |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (2.29)$$

Matrix Tensor Product:

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots \\ a_{21}B & a_{22}B & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \quad (2.30)$$

Example - CNOT Gate Construction:

CNOT (control c , target t) acts as:

$$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X \quad (2.31)$$

Interpretation: “If control is 0, apply identity to target. If control is 1, apply X to target.”

2.3 Dirac Notation (Bra-Ket Notation)

2.3.1 Definitions

Ket (Column Vector):

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (2.32)$$

Bra (Row Vector - Complex Conjugate Transpose):

$$\langle\psi| = (\alpha^* \quad \beta^*) \quad (2.33)$$

Bra-Ket (Inner Product):

$$\langle\psi|\phi\rangle = \alpha^* \gamma + \beta^* \delta \quad (2.34)$$

Ket-Bra (Outer Product):

$$|\psi\rangle \langle\phi| = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} (\gamma^* \quad \delta^*) = \begin{pmatrix} \alpha\gamma^* & \alpha\delta^* \\ \beta\gamma^* & \beta\delta^* \end{pmatrix} \quad (2.35)$$

2.3.2 Useful Identities

Completeness (Single Qubit):

$$|0\rangle \langle 0| + |1\rangle \langle 1| = I \quad (2.36)$$

This means any state can be decomposed:

$$|\psi\rangle = (|0\rangle \langle 0| + |1\rangle \langle 1|) |\psi\rangle = |0\rangle \underbrace{(\langle 0|\psi\rangle)}_{\alpha} + |1\rangle \underbrace{(\langle 1|\psi\rangle)}_{\beta} \quad (2.37)$$

Orthonormality:

$$\langle i|j\rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (2.38)$$

2.3.3 Common Bases

Computational Basis (Z-basis):

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (2.39)$$

Hadamard Basis (X-basis):

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (2.40)$$

Circular Basis (Y-basis):

$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (2.41)$$

Chapter 3

Quantum Mechanics Basics

3.1 The Five Postulates of Quantum Mechanics

3.1.1 Postulate 1: State Space

Theorem: State Space Postulate

The state of a quantum system is completely described by a unit vector $|\psi\rangle$ in a complex Hilbert space $\mathcal{H}$.

Mathematical Form (Single Qubit):

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (3.1)$$

Physical Interpretation:

$$|\alpha|^2 = \text{Probability of measuring } |0\rangle \quad (3.2)$$

$$|\beta|^2 = \text{Probability of measuring } |1\rangle \quad (3.3)$$

$$\text{Normalization } |\alpha|^2 + |\beta|^2 = 1 \text{ ensures total probability} = 1 \quad (3.4)$$

Example:

$$|\psi\rangle = \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} i |1\rangle \quad (3.5)$$

$$P(\text{measure } |0\rangle) = \left| \frac{1}{\sqrt{3}} \right|^2 = \frac{1}{3} \quad (3.6)$$

$$P(\text{measure } |1\rangle) = \left| \sqrt{\frac{2}{3}} i \right|^2 = \frac{2}{3} \quad (3.7)$$

$$\text{Total: } \frac{1}{3} + \frac{2}{3} = 1 \quad \checkmark \quad (3.8)$$

3.1.2 Postulate 2: Observables and Measurement Operators

Theorem: Observable Postulate

Every physical observable (measurable quantity) is represented by a Hermitian operator $\hat{O}$ with the property $\hat{O} = \hat{O}^\dagger$.

Key Properties:

- Eigenvalues of $\hat{O}$ are the possible measurement outcomes
- Eigenvectors form a complete orthonormal basis
- If system is in eigenstate $|v_i\rangle$: $\hat{O}|v_i\rangle = \lambda_i|v_i\rangle$, measurement gives λ_i with probability 1

Example - Computational Basis Measurement (Pauli Z):

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.9)$$

Eigenvalues: $\lambda_1 = +1$ (eigenstate $|0\rangle$), $\lambda_2 = -1$ (eigenstate $|1\rangle$)

Measurement returns: $+1$ if $|0\rangle$, or -1 if $|1\rangle$

Why Hermitian?

- **Real eigenvalues:** Measurement outcomes must be real numbers
- **Orthogonal eigenvectors:** Basis for complete measurement description

3.1.3 Postulate 3: Measurement and Collapse

Theorem: Measurement Postulate

When measuring observable $\hat{O}$ with eigenvectors $\{|v_i\rangle\}$ and eigenvalues $\{\lambda_i\}$ on state $|\psi\rangle$:

1. **Probability:** Probability of obtaining result λ_i is

$$P(\lambda_i) = |\langle v_i | \psi \rangle|^2 = |\langle v_i | \psi \rangle|^2 \quad (3.10)$$

2. **Collapse:** After measurement obtaining λ_i , state collapses to

$$|\psi\rangle_{\text{after}} = |v_i\rangle \quad (3.11)$$

Example - Measurement Collapse:

Initial state:

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle \quad (3.12)$$

Measure in computational basis (Z-basis):

$$P(0) = \frac{1}{3} \Rightarrow \text{collapse to } |0\rangle \quad (3.13)$$

$$P(1) = \frac{2}{3} \Rightarrow \text{collapse to } |1\rangle \quad (3.14)$$

Important: The original superposition is destroyed. We cannot recover the amplitudes after measurement.

3.1.4 Postulate 4: Time Evolution

Theorem: Schrödinger Equation

The time evolution of a closed quantum system is governed by:

$$i\hbar \frac{d|\psi\rangle}{dt} = \hat{H}|\psi\rangle \quad (3.15)$$

where $\hat{H}$ is the Hamiltonian operator and $\hbar = \frac{h}{2\pi}$ (reduced Planck constant).

Solution (Time-Independent Hamiltonian):

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle = U(t) |\psi(0)\rangle \quad (3.16)$$

where $U(t) = e^{-i\hat{H}t/\hbar}$ is the time evolution operator (unitary).

Application in Quantum Computing:

Quantum gates implement unitary operators. A gate applied for time t with Hamiltonian $\hat{H}$:

$$\text{Gate} = e^{-i\hat{H}t/\hbar} \quad (3.17)$$

Example - Rotation Gate:

The $R_x(\theta)$ gate (rotation around X-axis by angle θ):

$$R_x(\theta) = e^{-i\theta\hat{X}/2} \quad (3.18)$$

This comes from evolution with Hamiltonian $\hat{H} = \hat{X}$ for time $t = \theta/2$.

3.1.5 Postulate 5: Composite Systems

Theorem: Multipartite Systems

For a composite system of multiple subsystems, the Hilbert space is the tensor product:

$$\mathcal{H}_{\text{total}} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n \quad (3.19)$$

Example - Two-Qubit System:

Individual states: $|\psi_1\rangle = \alpha_1 |0\rangle + \beta_1 |1\rangle$ and $|\psi_2\rangle = \alpha_2 |0\rangle + \beta_2 |1\rangle$

Product state:

$$|\psi\rangle_{12} = |\psi_1\rangle \otimes |\psi_2\rangle = (\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) \quad (3.20)$$

Expanding:

$$|\psi\rangle_{12} = \alpha_1\alpha_2|00\rangle + \alpha_1\beta_2|01\rangle + \beta_1\alpha_2|10\rangle + \beta_1\beta_2|11\rangle \quad (3.21)$$

Important: Not all two-qubit states can be factored this way. Those that cannot are **entangled**.

3.2 Quantum States and Qubits

3.2.1 The Bloch Sphere

A single-qubit state can be parametrized by two real angles:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (3.22)$$

where $\theta \in [0, \pi]$ (polar angle) and $\phi \in [0, 2\pi]$ (azimuthal angle).

Bloch Sphere Representation:

$$x = \sin\theta \cos\phi \quad (3.23)$$

$$y = \sin\theta \sin\phi \quad (3.24)$$

$$z = \cos\theta \quad (3.25)$$

Key Points:

- North pole $(0, 0, 1)$: $|0\rangle$
- South pole $(0, 0, -1)$: $|1\rangle$
- Equator: Equal superpositions $|+\rangle, |-\rangle, |+i\rangle, |-i\rangle$
- Any rotation on Bloch sphere = unitary single-qubit operation

Note: Quantum computing is geometrically the art of rotating and entangling qubits on the Bloch sphere.

3.3 Superposition

3.3.1 Definition

Superposition is the fundamental quantum property allowing a system to exist in a linear combination of multiple eigenstates simultaneously until measurement forces it into one eigenstate.

3.3.2 Creating Superposition - Hadamard Gate

The Hadamard gate creates equal superposition from basis states:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (3.26)$$

Action:

$$H |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle \quad (3.27)$$

$$H |1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle \quad (3.28)$$

Derivation:

$$H |0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (3.29)$$

Probability Interpretation:

$$P(\text{measure } |0\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (3.30)$$

$$P(\text{measure } |1\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (3.31)$$

Equal probability of measuring either state!

3.3.3 Exponential Scaling

With n qubits all in superposition, we have 2^n simultaneous amplitudes.

Example - 3 Qubits:

$$(H \otimes H \otimes H) |000\rangle = \frac{1}{\sqrt{8}} \sum_{x=0}^7 |x\rangle \quad (3.32)$$

This single state represents all 8 possible 3-bit values simultaneously!

Classical vs Quantum:

Classical: To check 8 values, need 8 iterations (3.33)

Quantum: All 8 values in one superposition (quantum parallelism) (3.34)

This exponential scaling is the foundation of quantum speedup.

3.4 Entanglement

3.4.1 Definition

Entanglement occurs when a multi-qubit quantum system cannot be expressed as a tensor product of individual qubit states. The qubits are irreducibly correlated.

3.4.2 Separable vs Entangled States

Separable (Factorizable):

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = (\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) \quad (3.35)$$

Entangled:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (\text{Cannot be factored}) \quad (3.36)$$

Example - Bell State $|\Phi^+\rangle$:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (3.37)$$

Proof: that $|\Phi^+\rangle$ is Entangled (by contradiction):

Assume $|\Phi^+\rangle$ is separable:

$$|\Phi^+\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = (\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) \quad (3.38)$$

Expanding the right side:

$$(\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) = \alpha_1 \alpha_2 |00\rangle + \alpha_1 \beta_2 |01\rangle + \beta_1 \alpha_2 |10\rangle + \beta_1 \beta_2 |11\rangle \quad (3.39)$$

For this to equal $\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$:

$$\alpha_1 \alpha_2 = \frac{1}{\sqrt{2}} \quad \dots(1) \quad (3.40)$$

$$\alpha_1 \beta_2 = 0 \quad \dots(2) \quad (3.41)$$

$$\beta_1 \alpha_2 = 0 \quad \dots(3) \quad (3.42)$$

$$\beta_1 \beta_2 = \frac{1}{\sqrt{2}} \quad \dots(4) \quad (3.43)$$

From (2): Either $\alpha_1 = 0$ or $\beta_2 = 0$

Case 1: If $\alpha_1 = 0$, then from (1): $0 = \frac{1}{\sqrt{2}}$ Contradiction!

Case 2: If $\beta_2 = 0$, then from (3): $\beta_1 \cdot \alpha_2 = 0$

Sub-case 2a: $\beta_1 = 0$, then from (4): $0 = \frac{1}{\sqrt{2}}$ Contradiction!

Sub-case 2b: $\alpha_2 = 0$, then from (1): $\alpha_1 \cdot 0 = \frac{1}{\sqrt{2}}$ Contradiction!

All cases lead to contradiction. Therefore, $|\Phi^+\rangle$ **cannot be factored** and is **entangled**. $\square$

3.4.3 Bell States (Four Maximally Entangled States)

The four Bell states form an orthonormal basis for 2-qubit systems:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (\text{symmetric, even parity}) \quad (3.44)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \quad (\text{symmetric, odd parity}) \quad (3.45)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad (\text{antisymmetric, even parity}) \quad (3.46)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (\text{antisymmetric, odd parity}) \quad (3.47)$$

3.4.4 Creating Bell States

Circuit to create $|\Phi^+\rangle$:

```

|0  --[H]-- --
      |
|0  ----- --

```

Step-by-Step Evolution:

1. Initial state: $|00\rangle$
2. After Hadamard on first qubit:

$$(H \otimes I) |00\rangle = H |0\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \quad (3.48)$$

3. After CNOT (control=first, target=second):

$$\text{CNOT} \left[\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \right] = \frac{1}{\sqrt{2}}[\text{CNOT} |00\rangle + \text{CNOT} |10\rangle] \quad (3.49)$$

$$= \frac{1}{\sqrt{2}}[|00\rangle + |11\rangle] \quad (3.50)$$

$$= |\Phi^+\rangle \quad (3.51)$$

Explanation: - CNOT leaves $|00\rangle$ unchanged (control is 0, target unchanged)
 - CNOT flips target in $|10\rangle$ to $|11\rangle$ (control is 1, target flipped)

Result: Perfect correlation! If first qubit is 0, second is 0. If first qubit is 1, second is 1.

3.4.5 Non-Local Correlations

Once entangled, measuring one qubit instantaneously affects statistics of the other.

Example with $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

- Measure qubit 1 in computational basis
 - Get $|0\rangle$ with 50% probability $\Rightarrow$ qubit 2 is guaranteed to be $|0\rangle$
 - Get $|1\rangle$ with 50% probability $\Rightarrow$ qubit 2 is guaranteed to be $|1\rangle$
- The states are ****perfectly correlated****!

Important: This is not “spooky action at a distance” violating relativity.

- No information is transmitted faster than light
- Measurement results appear random to each observer individually
- Only when results are compared (at light speed) do we see correlation

Chapter 4

Quantum Gates and Circuits

4.1 Single-Qubit Gates

4.1.1 The Pauli Matrices

Pauli-X Gate (Quantum NOT)

Matrix:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (4.1)$$

Action on Basis States:

$$X |0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle \quad (4.2)$$

$$X |1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \quad (4.3)$$

Physical Interpretation: Flips qubit between $|0\rangle$ and $|1\rangle$ (quantum version of classical NOT).

Bloch Sphere: 180° rotation around X-axis.

Eigenanalysis:

$$\text{Eigenvalue } +1 : X |+\rangle = |+\rangle \quad \text{where } |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (4.4)$$

$$\text{Eigenvalue } -1 : X |-\rangle = -|-\rangle \quad \text{where } |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (4.5)$$

Proof:of Eigenvalue +1:

$$X |+\rangle = X \cdot \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}(X |0\rangle + X |1\rangle) = \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle) = |+\rangle = (+1) |+\rangle \quad \checkmark \quad (4.6)$$

Key Properties:

$$X^2 = I \quad (\text{applying } X \text{ twice is identity}) \quad (4.7)$$

$$X^\dagger = X \quad (X \text{ is Hermitian}) \quad (4.8)$$

$$\det(X) = -1 \quad (4.9)$$

$$\text{tr}(X) = 0 \quad (4.10)$$

Pauli-Y Gate**Matrix:**

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (4.11)$$

Action:

$$Y |0\rangle = i |1\rangle \quad (4.12)$$

$$Y |1\rangle = -i |0\rangle \quad (4.13)$$

Eigenanalysis:

$$\text{Eigenvalue } +1 : \quad |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i |1\rangle) \quad (4.14)$$

$$\text{Eigenvalue } -1 : \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i |1\rangle) \quad (4.15)$$

Properties:

$$Y^2 = I \quad (4.16)$$

$$Y^\dagger = Y \quad (4.17)$$

$$Y = iXZ \quad (4.18)$$

Pauli-Z Gate (Phase-Flip)**Matrix:**

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4.19)$$

Action:

$$Z |0\rangle = |0\rangle \quad (4.20)$$

$$Z |1\rangle = -|1\rangle \quad (4.21)$$

Physical Interpretation: Applies a ****phase flip**** to $|1\rangle$ only. Doesn't change measurement probability in computational basis, but affects interference.

Example - Effect on Superposition:

$$Z \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right] = \frac{1}{\sqrt{2}}(Z |0\rangle + Z |1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (4.22)$$

State transforms from $|+\rangle$ to $|-\rangle$ due to relative phase difference!

Eigenanalysis:

$$\text{Eigenvalue } +1 : |0\rangle \quad (4.23)$$

$$\text{Eigenvalue } -1 : |1\rangle \quad (4.24)$$

Properties:

$$Z^2 = I \quad (4.25)$$

$$Z^\dagger = Z \quad (4.26)$$

$$Z = -iXY \quad (4.27)$$

Pauli Matrix Relationships

Commutation Relations:

$$[X, Y] = XY - YX = 2iZ \quad (4.28)$$

$$[Y, Z] = YZ - ZY = 2iX \quad (4.29)$$

$$[Z, X] = ZX - XZ = 2iY \quad (4.30)$$

Anticommutation Relations:

$$\{X, Y\} = XY + YX = 0 \quad (4.31)$$

$$\{Y, Z\} = YZ + ZY = 0 \quad (4.32)$$

$$\{Z, X\} = ZX + XZ = 0 \quad (4.33)$$

Products:

$$XY = iZ, \quad YX = -iZ \quad (4.34)$$

$$YZ = iX, \quad ZY = -iX \quad (4.35)$$

$$ZX = iY, \quad XZ = -iY \quad (4.36)$$

4.1.2 The Hadamard Gate

Matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (4.37)$$

Action:

$$H |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle \quad (4.38)$$

$$H |1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle \quad (4.39)$$

Eigenanalysis:

$$\text{Eigenvalue } +1 : |+\rangle \quad (4.40)$$

$$\text{Eigenvalue } -1 : |-\rangle \quad (4.41)$$

MOST IMPORTANT PROPERTY: $H^2 = I$ **Proof::**

$$H^2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (4.42)$$

Matrix multiplication:

$$(1, 1) : 1 \cdot 1 + 1 \cdot 1 = 2 \quad (4.43)$$

$$(1, 2) : 1 \cdot 1 + 1 \cdot (-1) = 0 \quad (4.44)$$

$$(2, 1) : 1 \cdot 1 + (-1) \cdot 1 = 0 \quad (4.45)$$

$$(2, 2) : 1 \cdot 1 + (-1) \cdot (-1) = 2 \quad (4.46)$$

$$H^2 = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad \checkmark \quad (4.47)$$

Properties:

$$H^\dagger = H \quad (\text{Hermitian}) \quad (4.48)$$

$$H X H = Z \quad (\text{basis transformation}) \quad (4.49)$$

$$H Z H = X \quad (\text{basis transformation}) \quad (4.50)$$

4.1.3 Phase Gates**S Gate (Phase Gate)****Matrix:**

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad (4.51)$$

Action:

$$S |0\rangle = |0\rangle \quad (4.52)$$

$$S |1\rangle = i |1\rangle \quad (4.53)$$

Physical Interpretation: Applies 90° phase shift to $|1\rangle$.**Bloch Sphere:** 90° rotation around Z-axis.**KEY PROPERTY: $S^2 = Z$** **Proof::**

$$S^2 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z \quad \checkmark \quad (4.54)$$

Conjugate:

$$S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} \quad (4.55)$$

Property: $S^\dagger S = I$ (S is unitary)

T Gate (/8 Gate)

Matrix:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix} \quad (4.56)$$

Action:

$$T |0\rangle = |0\rangle \quad (4.57)$$

$$T |1\rangle = e^{i\pi/4} |1\rangle \quad (4.58)$$

Physical Interpretation: Applies 45° phase shift to $|1\rangle$.

FUNDAMENTAL PROPERTIES:

Property 1: $T^2 = S$

$$T^2 = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = S \quad \checkmark \quad (4.59)$$

Property 2: $T^4 = Z$ (Most Important!)

$$T^4 = (T^2)^2 = S^2 = Z \quad \checkmark \quad (4.60)$$

Or directly:

$$T^4 = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 0 & (e^{i\pi/4})^4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z \quad \checkmark \quad (4.61)$$

Property 3: $T^8 = I$

$$T^8 = (T^4)^2 = Z^2 = I \quad \checkmark \quad (4.62)$$

T has **order 8**: applying 8 times returns to identity.

T Dagger ($T^\dagger$):

$$T^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix} \quad (4.63)$$

Relation: $(T^\dagger)^4 = Z$ (same eigenvalue structure)

4.1.4 Rotation Gates

$R_x(\theta)$ - Rotation around X-Axis

Matrix:

$$R_x(\theta) = e^{-i\theta X/2} = \begin{pmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \quad (4.64)$$

Proof:using matrix exponential:

Using $X^2 = I$, we have:

$$e^{-i\theta X/2} = \sum_{n=0}^{\infty} \frac{(-i\theta/2)^n X^n}{n!} \quad (4.65)$$

Separating even and odd powers:

$$e^{-i\theta X/2} = \sum_{k=0}^{\infty} \frac{(-i\theta/2)^{2k}}{(2k)!} I + \sum_{k=0}^{\infty} \frac{(-i\theta/2)^{2k+1}}{(2k+1)!} X \quad (4.66)$$

$$= \cos(\theta/2) I - i \sin(\theta/2) X \quad (4.67)$$

$$= \cos(\theta/2) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i \sin(\theta/2) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \quad (4.68)$$

Action on $|0\rangle$:

$$R_x(\theta) |0\rangle = \cos(\theta/2) |0\rangle - i \sin(\theta/2) |1\rangle \quad (4.69)$$

Special Cases:

$$R_x(0) = I \quad (4.70)$$

$$R_x(\pi/2) |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i |1\rangle) \quad (4.71)$$

$$R_x(\pi) |0\rangle = -iX |0\rangle = -i |1\rangle \quad (4.72)$$

$R_y(\theta)$ - Rotation around Y-Axis

Matrix:

$$R_y(\theta) = e^{-i\theta Y/2} = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \quad (4.73)$$

Action on $|0\rangle$:

$$R_y(\theta) |0\rangle = \cos(\theta/2) |0\rangle + \sin(\theta/2) |1\rangle \quad (4.74)$$

Special Case:

$$R_y(\pi/2) |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle \quad (4.75)$$

Useful for creating equal superposition!

$R_z(\theta)$ - Rotation around Z-Axis**Matrix:**

$$R_z(\theta) = e^{-i\theta Z/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (4.76)$$

Action:

$$R_z(\theta) |1\rangle = e^{i\theta/2} |1\rangle \quad (4.77)$$

Relationships:

$$R_z(\pi/2) = S \quad (4.78)$$

$$R_z(\pi/4) = T \quad (4.79)$$

FUNDAMENTAL IDENTITY:

$$R_x(\theta) = H R_z(\theta) H \quad (4.80)$$

This shows X-rotation can be implemented using Z-rotation sandwiched between Hadamards!

4.1.5 Important Gate Relationships and Identities**Identity 1:** $HZH = X$ **Proof::**

$$HZ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4.81)$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \quad (4.82)$$

$$HZH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (4.83)$$

$$= \frac{1}{2} \begin{pmatrix} 1-1 & 1+1 \\ 1+1 & 1-1 \end{pmatrix} \quad (4.84)$$

$$= \frac{1}{2} \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \quad (4.85)$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X \quad \checkmark \quad (4.86)$$

Physical Interpretation: Hadamard basis transformation: measuring in Z-basis after Hadamard is equivalent to measuring in X-basis.

Identity 2: $HXH = Z$ **Proof::** Hermitian conjugate of Identity 1 (since $H^\dagger = H$).

Identity 3: $XY = iZ$ **Proof::**

$$XY = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = iZ \quad \checkmark \quad (4.87)$$

Identity 4: $YZ = iX$

$$YZ = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = iX \quad \checkmark \quad (4.88)$$

Identity 5: $ZX = -iY$

$$ZX = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = -iY \quad \checkmark \quad (4.89)$$

Chapter 5

Quantum Algorithms (Introduction)

5.1 Deutsch's Algorithm

5.1.1 Problem Definition

We are given a **black box** implementing function $f : \{0, 1\} \rightarrow \{0, 1\}$.

There are exactly four such functions:

Function	$f(0)$	$f(1)$	Type
f_1 (Identity)	0	1	Balanced
f_2 (NOT)	1	0	Balanced
f_3 (Constant-0)	0	0	Constant
f_4 (Constant-1)	1	1	Constant

Task: Determine if f is **constant** or **balanced** (and which type if possible).

Definition:

- **Constant:** $f(0) = f(1)$
- **Balanced:** $f(0) \neq f(1)$

5.1.2 Classical Solution

To determine if f is constant or balanced:

1. Query $f(0)$ to get 1 bit
2. Query $f(1)$ to get 1 bit
3. If results are same: constant. If different: balanced.

Total Queries: 2

5.1.3 Deutsch's Quantum Solution

Deutsch's algorithm solves this with only ****1 query****!

Algorithm Steps

Step 1: Initialize

$$|\psi_0\rangle = |01\rangle \quad (5.1)$$

Two qubits: first called query qubit, second called work qubit.

Step 2: Apply Hadamard to Both Qubits

$$(H \otimes H) |01\rangle = H |0\rangle \otimes H |1\rangle \quad (5.2)$$

$$H |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (5.3)$$

$$H |1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (5.4)$$

$$|\psi_1\rangle = \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle) = \frac{1}{2}[|00\rangle - |01\rangle + |10\rangle - |11\rangle] \quad (5.5)$$

Equivalently:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} \left[|0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) + |1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right] \quad (5.6)$$

Both qubits are now in superposition.

Step 3: Apply Quantum Oracle

The oracle implements:

$$U_f : |x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle \quad (5.7)$$

where $\oplus$ is XOR (addition mod 2).

Applying to our state:

$$|\psi_2\rangle = U_f |\psi_1\rangle = \frac{1}{2} \sum_{x \in \{0,1\}} [|x\rangle \otimes (U_f \left| \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\rangle)] \quad (5.8)$$

For each x :

$$U_f \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right] = \frac{1}{\sqrt{2}}(U_f |0\rangle - U_f |1\rangle) \quad (5.9)$$

$$= \frac{1}{\sqrt{2}}(|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle) \quad (5.10)$$

If $f(x) = 0$: $|0\rangle - |1\rangle$

If $f(x) = 1$: $|1\rangle - |0\rangle = -(|0\rangle - |1\rangle)$

So the work qubit remains $\frac{|0\rangle - |1\rangle}{\sqrt{2}}$, but the query qubit gets a phase:

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} [(-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle] \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (5.11)$$

****Phase Kickback in Action!**** The function values appear as phases on the query qubit.

Analysis by Cases:

Case 1: f is CONSTANT

If $f(0) = f(1) = 0$:

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} [|0\rangle + |1\rangle] \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |+\rangle \otimes |-\rangle \quad (5.12)$$

If $f(0) = f(1) = 1$:

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} [-|0\rangle - |1\rangle] \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} = -|+\rangle \otimes |-\rangle \quad (5.13)$$

Query qubit is in $|+\rangle$ state (up to global phase).

Case 2: f is BALANCED

If $f(0) = 0, f(1) = 1$:

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} [|0\rangle - |1\rangle] \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle \otimes |-\rangle \quad (5.14)$$

If $f(0) = 1, f(1) = 0$:

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} [-|0\rangle + |1\rangle] \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} = -|-\rangle \otimes |-\rangle \quad (5.15)$$

Query qubit is in $|-\rangle$ state (up to global phase).

Step 4: Apply Final Hadamard to Query Qubit

$$H|+\rangle = |0\rangle \quad (5.16)$$

$$H|-\rangle = |1\rangle \quad (5.17)$$

Proof::

$$H|+\rangle = H \cdot \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} (H|0\rangle + H|1\rangle) = \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) + \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right] = \frac{1}{2} [(|0\rangle + |1\rangle) + (|0\rangle - |1\rangle)] \quad (5.18)$$

Step 5: Measure Query Qubit

$$\text{Measurement result } 0 \Rightarrow f \text{ is } \mathbf{CONSTANT} \quad (5.19)$$

$$\text{Measurement result } 1 \Rightarrow f \text{ is } \mathbf{BALANCED} \quad (5.20)$$

5.1.4 Complete Circuit

```

|0  --[H]-- --[H]-- → Measure
          |          → Result: 0 if constant, 1 if balanced
|1  --[H]--Uf--

```

5.1.5 Quantum Advantage

Approach	Function Queries
Classical	2
Quantum (Deutsch)	1

Speedup Factor: 2x for single qubit, but exponential for n -qubit generalization (Deutsch-Jozsa algorithm)!

5.2 Quantum Parallelism

5.2.1 Concept

Quantum parallelism is the ability to evaluate a function on multiple inputs simultaneously using superposition.

5.2.2 Mathematical Formulation

For function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$:

$$|\text{output}\rangle = \sum_{x=0}^{2^n-1} |x\rangle |f(x)\rangle \quad (5.21)$$

This evaluates f on all 2^n inputs ****in a single step****!

5.2.3 Example: 3-Qubit Register

Starting with $|000\rangle$, apply H to all three qubits:

$$(H \otimes H \otimes H) |000\rangle = \frac{1}{\sqrt{8}} \sum_{x=0}^7 |x\rangle \quad (5.22)$$

This single quantum state simultaneously represents all 8 possible 3-bit values: $|000\rangle, |001\rangle, \dots, |111\rangle$.

Now apply oracle:

$$|\text{output}\rangle = \frac{1}{\sqrt{8}} \sum_{x=0}^7 |x\rangle |f(x)\rangle \quad (5.23)$$

All function values encoded!

5.2.4 Classical vs Quantum Computational Cost

Classical:

To compute $f(0), f(1), \dots, f(2^n - 1)$: Need 2^n separate function calls (5.24)

Quantum:

Using superposition: 1 superposed oracle call (5.25)

Apparent advantage: $2^n \times$ speedup!

5.2.5 The Measurement Problem

Limitation: If we measure directly, we get only ONE of the 2^n function values (with probability $1/2^n$ each).

To extract useful information from quantum parallelism requires:

- **Interference:** Amplitude amplification of correct answers
- **Entanglement:** Correlation between basis states
- **Clever algorithm design:** Like Deutsch's phase kickback

This is why quantum advantage is non-trivial—we must be clever about extracting information!

5.3 Exam Practice Questions with Solutions

5.3.1 Question 1: Gate Properties (5 marks)

Prove that $T^4 = Z$. Show all steps clearly.

Solution:

The T gate is defined as:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \quad (5.26)$$

Computing T^2 :

$$T^2 = T \cdot T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \quad (5.27)$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \cdot e^{i\pi/4} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} \quad (5.28)$$

Since $e^{i\pi/2} = \cos(\pi/2) + i \sin(\pi/2) = i$:

$$T^2 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = S \quad (5.29)$$

Now computing $T^4 = (T^2)^2 = S^2$:

$$S^2 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i \cdot i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i^2 \end{pmatrix} \quad (5.30)$$

Since $i^2 = -1$:

$$T^4 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z \quad \boxed{\checkmark} \quad (5.31)$$

Alternative direct method:

$$T^4 = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 0 & (e^{i\pi/4})^4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix} \quad (5.32)$$

Using Euler's formula: $e^{i\pi} = \cos \pi + i \sin \pi = -1 + 0 = -1$:

$$T^4 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z \quad \boxed{\checkmark} \quad (5.33)$$

5.3.2 Question 2: Hadamard Properties (6 marks)

Prove that $H^2 = I$ and $HZH = X$.

Solution:

Part (a): Prove $H^2 = I$

The Hadamard gate is:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (5.34)$$

Computing H^2 :

$$H^2 = H \cdot H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (5.35)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (5.36)$$

Computing each element:

$$(1, 1) \text{ element : } (1)(1) + (1)(1) = 2 \quad (5.37)$$

$$(1, 2) \text{ element : } (1)(1) + (1)(-1) = 0 \quad (5.38)$$

$$(2, 1) \text{ element : } (1)(1) + (-1)(1) = 0 \quad (5.39)$$

$$(2, 2) \text{ element : } (1)(1) + (-1)(-1) = 2 \quad (5.40)$$

Therefore:

$$H^2 = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad \boxed{\checkmark} \quad (5.41)$$

Part (b): Prove $HZH = X$

First compute ZH :

$$ZH = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (5.42)$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \quad (5.43)$$

Now compute HZH :

$$HZH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \quad (5.44)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \quad (5.45)$$

Computing elements:

$$(1, 1) : (1)(1) + (1)(-1) = 0 \quad (5.46)$$

$$(1, 2) : (1)(1) + (1)(1) = 2 \quad (5.47)$$

$$(2, 1) : (1)(1) + (-1)(-1) = 2 \quad (5.48)$$

$$(2, 2) : (1)(1) + (-1)(1) = 0 \quad (5.49)$$

$$HZH = \frac{1}{2} \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X \quad \boxed{\checkmark} \quad (5.50)$$

Physical Interpretation: Hadamard performs basis transformation. This identity shows that Z measurement in one basis equals X measurement in the other basis.

5.3.3 Question 3: Superposition and Measurement (5 marks)

A qubit is in state $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$.

(a) Verify this state is normalized.

(b) If measured in computational basis, what are the probabilities?

Solution:

Part (a): Normalization

A state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is normalized if $|\alpha|^2 + |\beta|^2 = 1$.

Here: $\alpha = \frac{1}{\sqrt{3}}, \beta = \sqrt{\frac{2}{3}}$

$$|\alpha|^2 + |\beta|^2 = \left| \frac{1}{\sqrt{3}} \right|^2 + \left| \sqrt{\frac{2}{3}} \right|^2 = \frac{1}{3} + \frac{2}{3} = 1 \quad \boxed{\checkmark} \quad (5.51)$$

Part (b): Measurement Probabilities

In computational basis $\{|0\rangle, |1\rangle\}$:

$$P(\text{measure } |0\rangle) = |\alpha|^2 = \frac{1}{3} \quad (5.52)$$

$$P(\text{measure } |1\rangle) = |\beta|^2 = \frac{2}{3} \quad (5.53)$$

Interpretation: If we perform this measurement on many identical copies, approximately 1/3 will yield |0 and 2/3 will yield |1 .

5.3.4 Question 4: Bell State Entanglement (6 marks)

Prove that the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is entangled.

Solution:

We prove by contradiction. Assume $|\Phi^+\rangle$ is separable (factorizable):

$$|\Phi^+\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \quad (5.54)$$

Let $|\psi_1\rangle = \alpha_1|0\rangle + \beta_1|1\rangle$ and $|\psi_2\rangle = \alpha_2|0\rangle + \beta_2|1\rangle$.

Then:

$$|\psi_1\rangle \otimes |\psi_2\rangle = (\alpha_1|0\rangle + \beta_1|1\rangle) \otimes (\alpha_2|0\rangle + \beta_2|1\rangle) \quad (5.55)$$

Expanding:

$$= \alpha_1\alpha_2|00\rangle + \alpha_1\beta_2|01\rangle + \beta_1\alpha_2|10\rangle + \beta_1\beta_2|11\rangle \quad (5.56)$$

For this to equal $|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$, we need:

$$\alpha_1\alpha_2 = \frac{1}{\sqrt{2}} \quad \dots(1) \quad (5.57)$$

$$\alpha_1\beta_2 = 0 \quad \dots(2) \quad (5.58)$$

$$\beta_1\alpha_2 = 0 \quad \dots(3) \quad (5.59)$$

$$\beta_1\beta_2 = \frac{1}{\sqrt{2}} \quad \dots(4) \quad (5.60)$$

From equation (2): Either $\alpha_1 = 0$ or $\beta_2 = 0$.

Case 1: If $\alpha_1 = 0$

From equation (1): $0 \cdot \alpha_2 = \frac{1}{\sqrt{2}}$, which gives $0 = \frac{1}{\sqrt{2}}$

This is a contradiction!

Case 2: If $\beta_2 = 0$

From equation (3): $\beta_1\alpha_2 = 0$, so either $\beta_1 = 0$ or $\alpha_2 = 0$.

- Sub-case 2a: $\beta_1 = 0$

From equation (4): $0 \cdot 0 = \frac{1}{\sqrt{2}}$, which gives $0 = \frac{1}{\sqrt{2}}$ Contradiction!

- Sub-case 2b: $\alpha_2 = 0$

From equation (1): $\alpha_1 \cdot 0 = \frac{1}{\sqrt{2}}$, which gives $0 = \frac{1}{\sqrt{2}}$ Contradiction!

All cases lead to contradiction. Therefore, $|\Phi^+\rangle$ cannot be factorized as a product state**.

Hence, $|\Phi^+\rangle$ is entangled.

5.3.5 Question 5: Deutsch's Algorithm (6 marks)

In Deutsch's algorithm, explain how the algorithm determines if function f is constant or balanced in ONE query. What is the role of phase kickback?

Solution:

Overview:

Deutsch's algorithm determines whether a single-bit function $f : \{0, 1\} \rightarrow \{0, 1\}$ is constant or balanced using only one query to the function oracle.

Key Insight - Phase Kickback:

The crucial mechanism is **phase kickback**. When the oracle is applied to a superposition of inputs, the function value appears as a **phase** on the query qubit, not as additional amplitude.

Detailed Explanation:

Step 1 - Preparation:

$$|\psi_0\rangle = |0\rangle_{\text{query}} \otimes |1\rangle_{\text{work}} \quad (5.61)$$

Step 2 - Create Superposition with Hadamards:

$$(H \otimes H)|\psi_0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (5.62)$$

Query qubit now represents BOTH inputs: $|0\rangle$ and $|1\rangle$ in superposition!

Step 3 - Apply Oracle (Phase Kickback):

The oracle implements: $|\psi\rangle_{\text{query}} \otimes |\phi\rangle_{\text{work}} \mapsto (-1)^{f(\psi)}|\psi\rangle_{\text{query}} \otimes |\phi\rangle_{\text{work}}$

The function value acts as a phase $(-1)^{f(x)}$:

$$\text{If } f(x) = 0 : (-1)^0 = +1 \quad (\text{no phase}) \quad (5.63)$$

$$\text{If } f(x) = 1 : (-1)^1 = -1 \quad (\text{phase of } \pi) \quad (5.64)$$

Analysis:

Constant Functions:

If $f(0) = f(1) = c$ (both 0 or both 1):

$$|\text{query}\rangle = \frac{1}{\sqrt{2}}[(-1)^c|0\rangle + (-1)^c|1\rangle] = (-1)^c \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = (-1)^c|+\rangle \quad (5.65)$$

Query qubit is in $|+\rangle$ state (up to global phase).

Balanced Functions:

If $f(0) \neq f(1)$:

$$|\text{query}\rangle = \frac{1}{\sqrt{2}}[\pm(|0\rangle \mp |1\rangle)] = \pm(|-\rangle) = |-\rangle \quad (5.66)$$

Query qubit is in $|-\rangle$ state (up to global phase).

Step 4 - Final Hadamard Distinguishes:

$$H|+\rangle = |0\rangle \Rightarrow \text{Measurement 0} \Rightarrow \text{CONSTANT} \quad (5.67)$$

$$H|-\rangle = |1\rangle \Rightarrow \text{Measurement 1} \Rightarrow \text{BALANCED} \quad (5.68)$$

Why This Works (Quantum Advantage):

- **Classical:** Must query $f(0)$ and $f(1)$ separately (2 queries minimum)
- **Quantum:** All information about both $f(0)$ and $f(1)$ encoded in **phases** of superposition
- **Single query** reveals pattern across ALL inputs through interference

Phase kickback is the quantum mechanism that enables exponential speedup

Appendix A: Gate Reference Sheet

Single-Qubit Gates

Gate	Matrix	Key Property	Order
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$X^2 = I$	2
Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	$Y^2 = I$	2
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$Z^2 = I$	2
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$H^2 = I$	2
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$S^2 = Z$	4
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	$T^4 = Z, T^8 = I$	8

Fundamental Identities

$$HZH = X \qquad HXH = Z \qquad (5.69)$$

$$XY = iZ \qquad YZ = iX \qquad ZX = iY \qquad (5.70)$$

$$S^2 = Z \qquad T^2 = S \qquad T^4 = Z \qquad (5.71)$$

$$H^2 = I \qquad X^2 = Y^2 = Z^2 = I \qquad (5.72)$$

$$R_x(\theta) = H R_z(\theta) H \qquad (5.73)$$

Appendix B: Formulas and Constants

Normalization and Probability

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1 \qquad (5.74)$$

$$P(\text{measure } |0\rangle) = |\alpha|^2, \quad P(\text{measure } |1\rangle) = |\beta|^2 \qquad (5.75)$$

Euler's Formula

$$e^{i\theta} = \cos \theta + i \sin \theta \qquad (5.76)$$

Inner Product and Orthonormality

$$\langle \psi | \phi \rangle = \alpha^* \gamma + \beta^* \delta, \quad \langle i | j \rangle = \delta_{ij} \quad (5.77)$$

Tensor Product

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix} \psi_1 \phi_1 \\ \psi_1 \phi_2 \\ \psi_2 \phi_1 \\ \psi_2 \phi_2 \end{pmatrix} \quad (5.78)$$